

09.2024

PRACTICAL-1

SLIDING WINDOW

AIM:-

To write a program to implement flow control at data link layer using SLIDING WINDOW PROTOCOL and to simulate the flow of frames from one node to another.

CODE:-

Sender.py

import time

window_size = int(input("Enter window size:"))

message = input("Enter the message:")

frames = []

for i in range(0, len(message), 5):

data = message[i:i+5]

frames.append([i//5, data])

print("\n Frames created:")

for frame in frames:

print("Frame No:", frame[0], "Data:", frame[1])

start = 0

while start < len(frames):

end = min(start + window_size, len(frames))

print("\n Sending frames:..")

with open("Sender-Buffer.txt", "w") as file:

for i in range(start, end):

file.write(f"{frames[i][0]} {frames[i][1]}\n")

print("Sending Frame", frames[i][0], ":", frames[i][1])

print("\n Waiting for ACK...")

time.sleep(2)

try:

with open("Receiver - Buffer.txt", "r") as file:

ack = int(file.read().strip())

print("ACK received:", ack)

except:

print("No ACK received")

continue

if ack == end - 1:

print("ACK is correct.")

start = end

else:

print("NACK / Wrong ACK!")

print("Resending frames from frame", ack + 1)

if ack >= start:

start = ack + 1

else:

start = start

print("\n All frames transmitted successfully!")

Receiver.py

```
import time
import os
print("Receiver started...")
expected-frame = 0
old-buffer = ""
while True:
    if not os.path.exists("Sender-Buffer.txt"):
        time.sleep(1)
        continue
    with open("Sender-Buffer.txt", "r") as file:
        data = file.read()
    if data == old-buffer:
        time.sleep(1)
        continue
    old-buffer = data
    lines = data.strip().split("\n")
    print("\n Received Frames:")
    last-ack = expected-frame - 1
    for line in lines:
        parts = line.split(",")
        frame-no = int(parts[0])
        frame-data = parts[1]
        print("Frame", frame-no, ":", frame-data)
        if frame-no == expected-frame:
            print("Frame accepted")
            last-ack = frame-no
            expected-frame += 1
        else:
            print("Wrong Frame!")
            break
    print("Sending ACK:", last-ack)
    with open("Receiver-Buffer.txt", "w") as file:
        file.write(str(last-ack))
    last-frame = int(lines[-1].split(",")[0])
    if last-ack == last-frame:
        print("\n All frames received successfully!")
        break
    time.sleep(1)
```

OUTPUT:-

> python Sender.py

Enter window size: 3

Enter the message: computer Networks

Frames created:

Frame No: 0 Data: compu

Frame No: 1 Data: ter N

Frame No: 2 Data: etwork

Frame No: 2 Data: ds

Sending frames...

sending Frame 0 : Compu

sending Frame 1 : ter N

sending Frame 2 : etuor

Waiting for ACK...

ACK received : 3

NACK / Wrong ACK!

Resending frames from Frame 4

> python Receiver.py

Receiver started...

Received frames:

Frame 0 : Compu

Frame accepted

Frame 1 : ter N

Frame accepted

Frame 2 : etuor

Frame accepted

Sending ACK : 2

All frames received successfully!

Elk

RESULT :-

The sliding window protocol was successfully implemented using Python. Frames were transmitted and acknowledged successfully, demonstrating reliable data transmission.